

- a) What are vanishing and exploding gradients? How do they affect neural networks? Explain in detail.

Vanishing and exploding gradients are problems that can occur while training deep neural networks during backpropagation. During backpropagation, gradients are repeatedly multiplied as they move from the output layer toward the earlier layers.

Vanishing gradients occur when these gradients become extremely small, approaching zero. As a result, the earlier layers receive very small weight updates and learn very slowly or stop learning. This commonly occurs in very deep networks and with activation functions such as sigmoid and tanh.

Exploding gradients occur when gradients become extremely large. This causes very large changes to the weights, making training unstable and potentially causing the loss to become extremely large or NaN.

These problems can be reduced using proper weight initialization, ReLU-type activation functions, batch normalization, gradient clipping, and residual/skip connections. ResNet50 uses residual connections, which provide a better path for gradients to flow through deep networks and help make deep network training more stable.

- b) Explain the role of padding and stride in convolutional layers? How will you decide when to use the padding.

Padding uses one extra layer in image so that corner elements also get importance. When kernel is iterated over the image corner elements also get importance as compared to middle elements. Padding helps to extract more information.

Padding is used when there is important information on edges.

- c) How does data augmentation help in improving CNN performance?

Data augmentation helps to train the model in a better way because after data augmentation model gets much wider training data so it can learn spatial hierarchies in a much better way.